THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** hydrochloric acid	HCl helps digestion
2	**(d)** 10	Basic: pH > 7
3	**(a)** dry slaked lime $Ca(OH)_2$	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$
4	**(b)** Incorrect — only water-soluble bases are alkalis	e.g. $Cu(OH)_2$ is a base but insoluble
5	**(a)** Student A is correct	Neutral
6	**(c)** $Na_2CO_3 \cdot 10\,H_2O$	Decahydrate
7	**(c)** hydrogen	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$
8	**(c)** 5.5	Enamel corrodes below 5.5
9	**(d)** sodium hydrogencarbonate	Baking soda
10	**(a)** H_3O^+	Hydronium ions

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acidic behaviour needs H^+ ions, which are produced only in the presence of water; dry HCl does not ionise.	2
12	Calcium sulphate hemihydrate, $CaSO_4 \cdot \frac{1}{2}H_2O$	1+1
13	They contain acids which react with the metal to form toxic compounds, spoiling the food.	2
14	False — dilution decreases the concentration of H_3O^+ ions per unit volume.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Acid + base → salt + water, e.g. $NaOH + HCl \longrightarrow NaCl + H_2O$; antacid relieving indigestion / baking soda on an ant sting	1+1+1
16	$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$; uses: baking powder, antacid, soda-acid fire extinguishers	1+1+1
17	(i) D (ii) C (iii) B (iv) A (v) E. Increasing H^+: C < E < D < A < B	2+1
	OR: Electrolysis of brine: $2\,NaCl(aq) + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$; chlorine at anode, hydrogen at cathode, NaOH near cathode	1+1+1
18	Fixed number of water molecules in a crystal. $CuSO_4 \cdot 5\,H_2O$ loses water on heating and becomes white anhydrous $CuSO_4$; turns blue again with water.	1+1+1
	OR: Acid rain lowers the pH of river water; aquatic animals survive only in a narrow pH range (about 7–7.8) close to neutral.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) salt + water + CO_2: $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (ii) turns lime water milky: $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (iii) milkiness disappears: $CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$	2+2+1
	OR: (i) Heating gypsum at 373 K: $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}H_2O + \frac{3}{2}H_2O$ (ii) sets into hard gypsum (iii) plaster for fractured bones, toys, decoration, smooth surfaces (iv) it absorbs moisture and sets hard	2+1+1+1
20	(i) methanoic (formic) acid (ii) it is a mild base which neutralises the acid (iii) neutralisation (iv) more than 7 (v) calamine solution / dock plant leaves for nettle	1+1+1+1+1
	OR: (i) Strong: ionise completely (HCl); weak: partly (CH_3COOH) (ii) mixture of indicators showing different colours at different pH (iii) (a) stomach HCl; excess acid neutralised by antacids (b) plants need a specific pH; acidic soil treated with lime	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** onion	Smell changes in acid/base
2	**(d)** pink	Pink in bases
3	**(b)** basic	Metal oxides are basic
4	**(b)** Incorrect — less reactive metals like copper do not give hydrogen	Cu does not displace H
5	**(b)** Student A is correct	Lactic acid forms
6	**(b)** calcium oxychloride	$CaOCl_2$
7	**(c)** oxalic acid	NCERT table
8	**(b)** strongly acidic	Very low pH
9	**(d)** highly exothermic	So acid is added to water slowly
10	**(c)** 7	e.g. NaCl is neutral

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The process is highly exothermic; adding water to acid can make the mixture splash and the glass may break. Adding acid slowly to water keeps it safe.	2
12	$Na_2CO_3 \cdot 10\,H_2O$; used in glass, soap and paper industries / removing permanent hardness of water / cleaning	1+1
13	It contains ions (H^+/H_3O^+ and anions) which carry current.	2
14	False — they do not ionise to give H^+ ions in water, so they do not conduct electricity and are not acidic.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (ii) $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (iii) $2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$	1+1+1
16	Baking soda + mild edible acid (tartaric acid). On heating, CO_2 is released which makes the cake rise. With only baking soda, sodium carbonate forms, giving a bitter taste.	1+1+1
17	Scale 0–14 to measure strength of acids/bases; higher H^+ concentration → lower pH; quicklime neutralises acidic soil.	1+1+1
	OR: Bleaching powder: bleaching cotton/linen, disinfecting drinking water, oxidising agent. Baking soda: baking powder, antacid, fire extinguishers.	1½+1½
18	pH decreases (lactic acid forms). Baking soda makes milk slightly alkaline so it takes longer to set into curd.	1+2
	OR: (i) acid turns it red; base no change (ii) base turns it blue; acid no change (iii) acid — red, base — yellow	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) bleaching powder (ii) chlorine on dry slaked lime: $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (iii) bleaching in textile/paper industry, disinfecting water, oxidising agent	1+2+2
	OR: (i) Zinc granules + dilute H_2SO_4 in a test tube; gas passed through soap solution forms bubbles (ii) $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (iii) burning candle near bubble — pop sound	3+1+1
20	(i) 5.5 (ii) calcium hydroxyapatite (calcium phosphate) (iii) they degrade sugar and food particles producing acids (iv) neutralises excess acid (v) basic	1+1+1+1+1
	OR: (i) electrolysis of brine; products are chlorine and alkali (NaOH) (ii) $2\,NaCl\,(aq) + 2\,H_2O\,(l) \longrightarrow 2\,NaOH\,(aq) + Cl_2\,(g) + H_2\,(g)$ (iii) H_2: fuels, margarine; Cl_2: water treatment, PVC; NaOH: soaps, degreasing	1+1+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.